

COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 6

6 Computation Theory (amp12)

- Lambda calculus (a) Define the Church numerals for zero ($\underline{0}$), one ($\underline{1}$) and for an arbitrary natural number ($\underline{n}$). [2 marks]

Answer: $\underline{0} \triangleq \lambda f x. x$, $\underline{1} \triangleq \lambda f x. f x$ and in general the n th Church numeral is $\underline{n} \triangleq \lambda f x. f^n x$, where $M^n N$ is defined by:
$$\begin{cases} M^0 N &= N \\ M^{n+1} N &= M(M^n N) \end{cases}$$

- (b) Define encodings of Booleans as λ -terms (**True**, **False** and **If**). [1 mark]

Answer: **True** $\triangleq \lambda x y. x$, **False** $\triangleq \lambda x y. y$ and **If** $\triangleq \lambda f x y. f x y$.

- (c) Explain what it means for a λ term to *represent* a number-valued partial function of n numerical arguments; do the same for one returning Boolean instead of numerical results. [3 marks]

Answer: A λ -term F represents $f : \mathbb{N}^n \rightarrow \mathbb{N}$ if for all $(x_1, \dots, x_n) \in \mathbb{N}^n$ and $y \in \mathbb{N}$,

- if $f(x_1, \dots, x_n)$ is defined and equal to y , then $F \underline{x_1} \dots \underline{x_n}$ is β -equal to $\underline{y}$
- if $f(x_1, \dots, x_n)$ is undefined, then $F \underline{x_1} \dots \underline{x_n}$ has no β -normal form.

The definition for a Boolean-valued partial function is the same except that we let y range over Booleans $\{0, 1\}$ and in the first clause we require $F \underline{x_1} \dots \underline{x_n}$ to be β -equal to **True** (respectively **False**) when $y = 1$ (respectively $y = 0$).

- (d) Give λ -terms that represent the following functions:

- (i) successor (**Succ**) [1 mark]

Answer: **Succ** $\triangleq \lambda n f x. f (n f x)$

- (ii) test for zero (**Eq₀**) [1 mark]

Answer: **Eq₀** $\triangleq \lambda x. x(\lambda y. \text{False}) \text{True}$

- (e) Define encodings of pairing and projections (**Pair**, **Fst** and **Snd**). [2 marks]

Answer: **Pair** $\triangleq \lambda x y f. f x y$, **Fst** $\triangleq \lambda f. f \text{True}$ and **Snd** $\triangleq \lambda f. f \text{False}$.

- (f) What function $\mathbb{N} \rightarrow \mathbb{N}$ is represented by the following λ -term? Carefully justify your answer.

$\lambda x. \text{Snd}(x(\lambda y. \text{Pair}(\text{Succ}(\text{Fst } y))(\text{Fst } y))(\text{Pair } \underline{0} \underline{0}))$

[6 marks]

Answer: **Claim:** the λ -term represents the predecessor function pred , whose value at 0 is 0 and whose value at $n + 1$ is n .

Define $P \triangleq \lambda y. \text{Pair}(\text{Succ}(\text{Fst } y))(\text{Fst } y)$. Then one can prove the claim by showing by induction on n that

$$\text{Fst}(\underline{n} P (\text{Pair } \underline{0} \underline{0})) =_{\beta} \underline{n} \quad (1_n)$$

$$\text{Snd}(\underline{n} P (\text{Pair } \underline{0} \underline{0})) =_{\beta} \underline{\text{pred } n} \quad (2_n)$$

Base case:

$$\begin{aligned} \text{Fst}(\underline{0} P (\text{Pair } \underline{0} \underline{0})) &=_{\beta} \text{Fst}(\text{Pair } \underline{0} \underline{0}) =_{\beta} \underline{0} \\ \text{Snd}(\underline{0} P (\text{Pair } \underline{0} \underline{0})) &=_{\beta} \text{Snd}(\text{Pair } \underline{0} \underline{0}) =_{\beta} \underline{0} =_{\beta} \underline{\text{pred } 0} \end{aligned}$$

Induction step: assuming (1_n) , then we have

$$\begin{aligned} \text{Fst}(\underline{n+1} P (\text{Pair } \underline{0} \underline{0})) &=_{\beta} \text{Fst}(P(\underline{n} P (\text{Pair } \underline{0} \underline{0}))) \\ &=_{\beta} \text{Fst}(\text{Pair}(\text{Succ}(\text{Fst}(\underline{n} P (\text{Pair } \underline{0} \underline{0}))))(\text{Fst}(\underline{n} P (\text{Pair } \underline{0} \underline{0}))) \\ &=_{\beta} \text{Succ}(\text{Fst}(\underline{n} P (\text{Pair } \underline{0} \underline{0}))) \\ &=_{\beta} \text{Succ } \underline{n} \quad \text{by } (1_n) \\ &=_{\beta} \underline{n+1} \end{aligned}$$

and

$$\begin{aligned} \text{Snd}(\underline{n+1} P (\text{Pair } \underline{0} \underline{0})) &=_{\beta} \text{Snd}(P(\underline{n} P (\text{Pair } \underline{0} \underline{0}))) \\ &=_{\beta} \text{Snd}(\text{Pair}(\text{Succ}(\text{Fst}(\underline{n} P (\text{Pair } \underline{0} \underline{0}))))(\text{Fst}(\underline{n} P (\text{Pair } \underline{0} \underline{0}))) \\ &=_{\beta} \text{Fst}(\underline{n} P (\text{Pair } \underline{0} \underline{0})) \\ &=_{\beta} \underline{n} \quad \text{by } (1_n) \\ &=_{\beta} \underline{\text{pred}(n+1)} \end{aligned}$$

so that both (1_{n+1}) and (2_{n+1}) hold.

- (g) Give with justification a λ -term that represents the function mapping each pair of numbers (m, n) to **True** if $m \leq n$ and to **False** otherwise. [Hint: use the λ -terms from parts (d)(ii) and (f).] [4 marks]

Answer: $\text{Leq} \triangleq \lambda x y. \text{Eq}_0(y \text{Pred } x)$, where Pred is the λ -term from part (f).

(Because then

$$\text{Leq } \underline{m} \underline{n} =_{\beta} \text{Eq}_0(\underline{n} \text{Pred } \underline{m}) =_{\beta} \text{Eq}_0(\text{Pred}^n \underline{m}) =_{\beta} \text{Eq}_0(\underline{m} \dot{-} n) =_{\beta} \begin{cases} \text{True} & \text{if } m \leq n \\ \text{False} & \text{otherwise.} \end{cases}$$
